



Adapting NASA F Prime with FreeRTOS for STM32-Based Embedded Flight Software

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Introduction

Small satellite missions increasingly depend on software-defined behavior to coordinate subsystem health, command handling, telemetry, fault response, and mission operations. For CubeSTEP-Cerberus, the flight software targets deployment on the EnduroSat On-Board Computer Type I, based on the STM32H753 microcontroller, while supporting a thermal experiment payload under strict power, communications, memory, and timing constraints.

NASA JPL’s F Prime framework provides a reusable, component-based architecture for flight software development. However, many F Prime deployments assume a Linux or POSIX-like execution environment. This project adapts F Prime for FreeRTOS-based STM32 embedded platforms, with emphasis on the EnduroSat OBCI flight computer using the STM32H753 microcontroller.

The work demonstrates that F Prime can be extended to previously unsupported bare-metal and RTOS-based targets through FreeRTOS Operating System Abstraction Layer (OSAL) modifications, removal of POSIX assumptions, board-specific initialization, custom toolchain support, and deployment-oriented validation on STM32 hardware.

OSAL Adaptation & Embedded Challenges

Porting F Prime to STM32 FreeRTOS is primarily a deployment and hardware integration problem. The OSAL must translate F Prime execution assumptions into deterministic FreeRTOS primitives while respecting STM32 memory, stack, and interrupt constraints.

Task execution	F’ active components are mapped to FreeRTOS tasks with explicit priorities and stack sizes.
Memory control	Allocation is aligned with the STM32H753 memory map, FreeRTOS heap configuration, startup files, and linker script.
Synchronization	FreeRTOS queues, mutexes, and semaphores preserve component execution semantics and protect shared resources.
Non-POSIX operation	Filesystem, process, standard I/O, and host-timing assumptions are isolated or removed for the 32-bit microcontroller target.
Board bring-up	Clock, GPIO, UART, I2C, SPI, memory-region, and interrupt initialization occur before F’ topology startup on the EnduroSat OBCI.

System Architecture

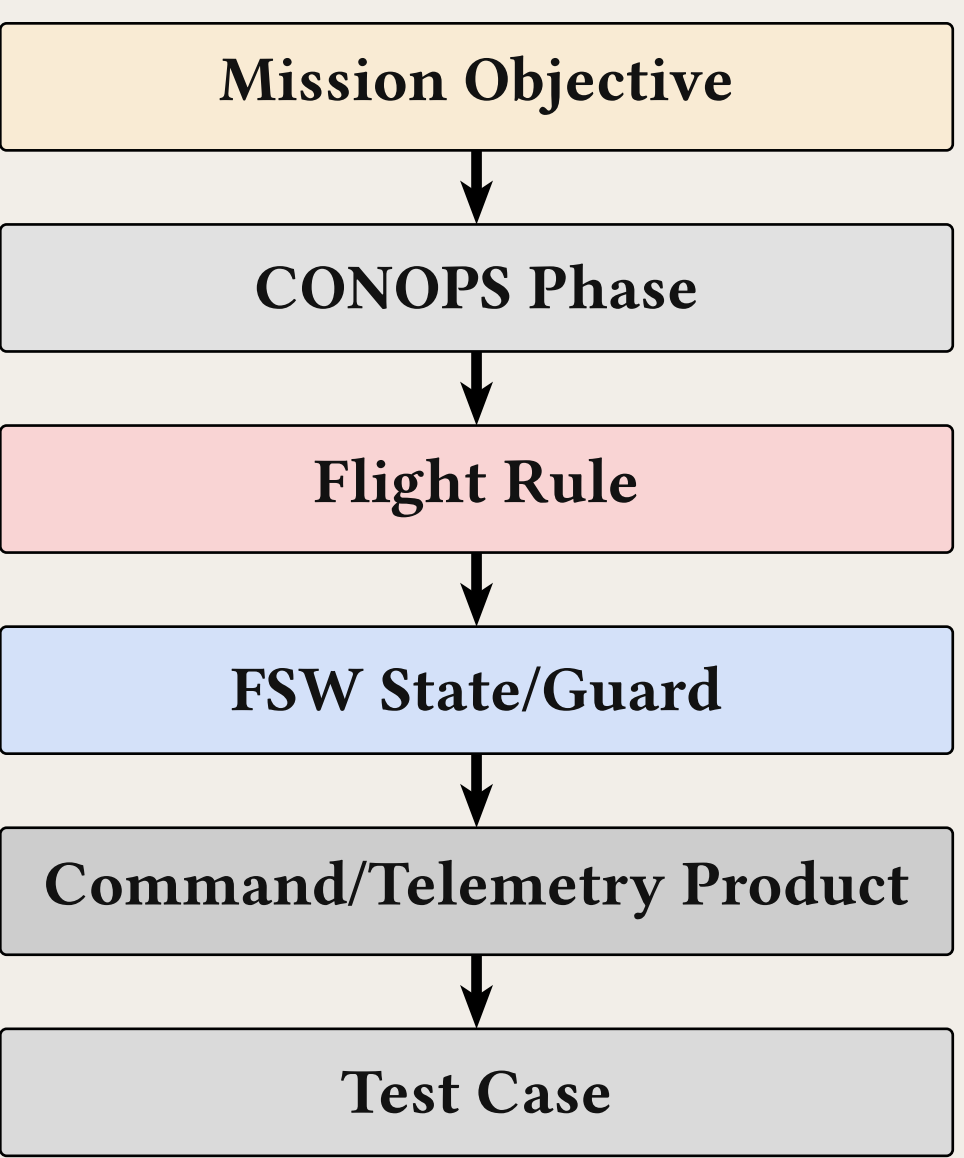
Initialization state is entered after deploying from the launcher. Detumbling executes until reaching stable body rates, followed by the deployment of the antenna and solar panels. The spacecraft then sends its first beacon for ground station connection. During this state, early subsystem setup takes place.

Safe state is entered whenever the spacecraft is undergoing degraded health. The spacecraft operates with minimum required functionality regarding power, communication, and recoverability. Once the spacecraft returns to acceptable conditions, a transition is made into idle. Commands can still be queued, but execution is deferred until transitioning into the appropriate state.

Idle state is the nominal state, performing health checks, peripheral polling, memory scrubbing, and ground-contact monitoring. Experiment commands can be queued, but execution is delayed until guards for the experiment state are satisfied.

Experiment state is entered upon satisfying requirements for mission phase, resource state, payload readiness, and data-storage. This state contains setup, execution, and closeout substates. Once operator inputs for OHP variant, sunlight phase, and heater power level are verified, and once the temperature, power margin, attitude stability, and storage availability meet requirements, a transition to execution is made for thermal data collection. At closeout, data is stored and a transition to idle is made.

Transmit state



Current Progress & Results

EnduroSatOBC	validated no-boot Fprime/FreeRTOS proof of concept on the OBC
GDS Connectivity	Implimented a register-level UART driver and confirmed F’ GDS communication through the F’ framing stack.
Additional Issues	Resolved memory allocation issues in upstream F Prime OSAL FreeRTOS implementation

Current progress has validated the practicality of using F Prime on the endurostat OBCI. We are now able to evaluate system viability with next steps focusing on refining implementation and software component integration

F’+FreeRTOS Implimentation

F Prime components are mapped onto FreeRTOS execution primitives while preserving command, telemetry, event, and scheduling behavior.

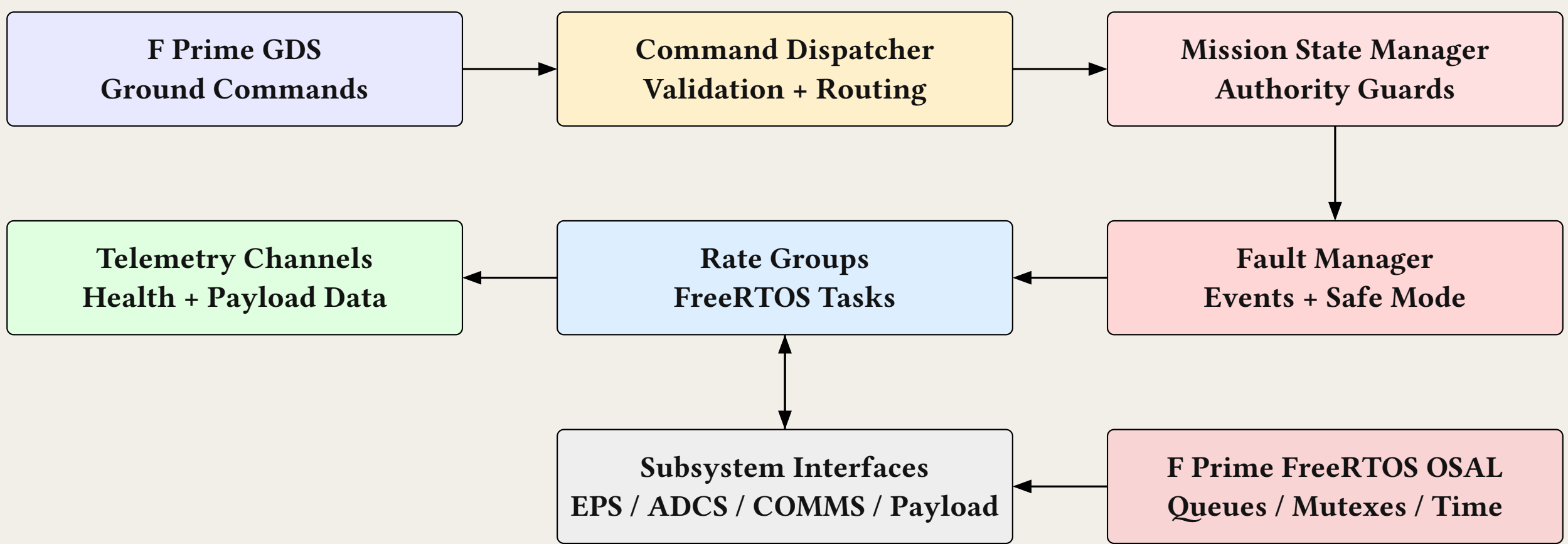


Figure 2: Command, Telemetry, Scheduling, and Fault Flow

Command handling is routed through F Prime command dispatchers and checked against mission-state authority rules before reaching subsystem components. Telemetry and events flow back through F Prime channels for downlink through the Ground Data System. Periodic behavior is driven by F Prime rate groups backed by FreeRTOS tasks, queues, and synchronization primitives.

Future Work and Flight Readiness

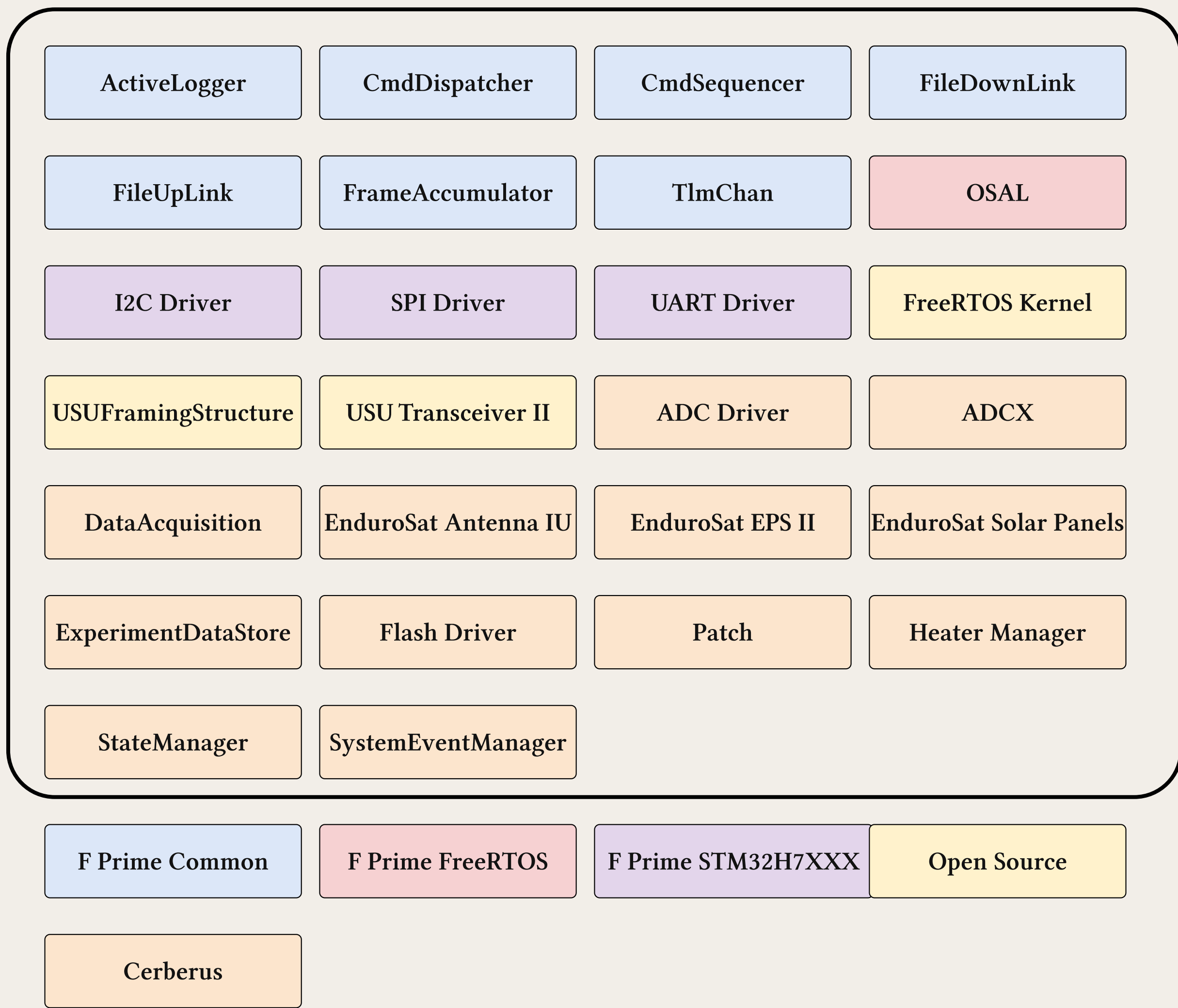
Next steps focus on converting the deployment from a successful bring-up into a flight-ready software baseline.

- Expand subsystem interface components for EPS, ADCS, communications, payload heaters, thermocouples, and nonvolatile storage.
- Complete software-in-the-loop and hardware-in-the-loop tests for nominal operations and representative fault cases.
- Add long-duration runtime testing to characterize stack usage, heap stability, timing jitter, queue behavior, and telemetry throughput.
- Formalize mission-stage constraints as testable requirements tied to command authorization and transition guards.
- Integrate experiment scheduling with communications-window planning and onboard data-volume limits.

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Software Architecture



Conclusion

This work demonstrates that F Prime can be extended beyond conventional Linux/POSIX-style deployments to constrained STM32/FreeRTOS spacecraft hardware. By adapting the OSAL, board initialization, memory configuration, and execution model, the project preserves F Prime’s command, telemetry, scheduling, and fault-management structure while enabling deployment on the EnduroSat OBCI pathway.

The current result is a functioning embedded F Prime/FreeRTOS baseline for CubeSTEP-Cerberus. The next step is to convert this bring-up into a flight-ready software baseline through subsystem integration, software-in-the-loop testing, hardware-in-the-loop testing, and long-duration runtime characterization.

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